Assignment: Library Management System (Python OOP)

Title

Library Management System using Python OOP

Objective

Develop a console-based Library Management System using Object-Oriented Programming (OOP) concepts in Python. The project should demonstrate your understanding of classes, objects, constructors, inheritance, encapsulation, properties, class methods, static methods, and composition.

Learning Outcomes

After completing this assignment, students should be able to:

- Create and use classes and objects
- Work with constructors (__init__)
- Use instance variables and methods
- Use class variables and class methods
- Create static methods
- Apply encapsulation using private attributes
- Use @property for getter and setter
- Implement inheritance and method overriding
- Apply composition between classes
- Organize Python projects into multiple files
- Push a complete project to GitHub

Project Description

A library wants to replace its paper-based system with a simple software application.
The system should allow a librarian to:

- Add new books
- Register members
- Borrow books
- Return books
- View all books
- View all members
- Search books by title
- Exit the program

All data will remain in memory (database is not required).

Required Classes

1. Person (Parent Class)

Attributes
name
age

Methods

display_info()

2. Member (Child of Person)
Additional Attributes

member_id
borrowed_books

Methods

borrow_book()
return_book()
display_info() (Override)

3. Book
Attributes
title
author
isbn
available

Methods

display_book()

Use encapsulation for available.

Create
Getter
Setter

Read-only property if needed

4. Library

Attributes

books
members

Methods

add_book()
register_member()
borrow_book()
return_book()
show_books()
show_members()
search_book()

Use composition, because a Library contains Books and Members.

Sample Input / Output

```
==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Search Book
8. Exit

Enter your choice: 1

----- Add New Book -----
Enter Book Title : Python Programming
Enter Author     : John Smith
Enter ISBN       : B101

Book added successfully!

Press Enter to continue...
```

```
==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Show All Books
6. Show All Members
7. Search Book
8. Exit

Enter your choice: 1

----- Add New Book -----
Enter Book Title : Data Structures
Enter Author     : Ahmed Ali
Enter ISBN       : B102

Book added successfully!

Press Enter to continue...


==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 2

----- Register Member -----

Enter Member ID : M001
Enter Name      : Naimur
Enter Age       : 23

Member registered successfully!

Press Enter to continue...


==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 2
```

```
----- Register Member -----

Enter Member ID : M002
Enter Name      : Sara
Enter Age       : 21

Member registered successfully!

Press Enter to continue...



==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 5

------------- BOOK LIST -------------

ISBN    : B101
Title   : Python Programming
Author  : John Smith
Status  : Available

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ISBN    : B102
Title   : Data Structures
Author  : Ahmed Ali
Status  : Available

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Press Enter to continue...



==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 6

----------- MEMBER LIST ------------

Member ID      : M001
Name           : Naimur
```

```
Age            : 22
Borrowed Books  : 0

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Member ID      : M002
Name           : Sara
Age            : 21
Borrowed Books  : 0

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Press Enter to continue...



==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 3

------ Borrow Book ------

Enter Member ID : M001
Enter Book ISBN : B101

Book borrowed successfully.

Press Enter to continue...



==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 5

------------- BOOK LIST -------------

ISBN    : B101
Title   : Python Programming
Author  : John Smith
Status  : Borrowed

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```

ISBN : B102
Title : Data Structures
Author : Ahmed Ali
Status : Available

Press Enter to continue...

==
LIBRARY MANAGEMENT SYSTEM
==

Enter your choice: 3

------ Borrow Book ------

Enter Member ID : M002
Enter Book ISBN : B101

Sorry! This book is currently unavailable.

Press Enter to continue...

==
LIBRARY MANAGEMENT SYSTEM
==

Enter your choice: 7

------ Search Book ------

Enter Book Title : Python Programming

Book Found!

ISBN : B101
Title : Python Programming
Author : John Smith
Status : Borrowed

Press Enter to continue...

```
==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 4

------ Return Book ------

Enter Member ID : M001
Enter Book ISBN : B101

Book returned successfully.

Press Enter to continue...


==========================================
     LIBRARY MANAGEMENT SYSTEM
==========================================

Enter your choice: 5

------------- BOOK LIST -------------

ISBN    : B101
Title   : Python Programming
Author  : John Smith
Status  : Available

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ISBN    : B102
Title   : Data Structures
Author  : Ahmed Ali
Status  : Available

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Press Enter to continue...
```

Example Error Cases

Duplicate ISBN

```
Enter your choice: 1
```

Enter Book Title : Python Programming
Enter Author : John Smith
Enter ISBN : B101

Error: ISBN already exists.

Duplicate Member ID

Enter your choice: 2

Enter Member ID : M001
Enter Name : Hasan
Enter Age : 20

Error: Member ID already exists.

Invalid Member

Enter your choice: 3

Enter Member ID : M999
Enter Book ISBN : B101

Member not found.

Book Not Found

Enter your choice: 3

Enter Member ID : M001
Enter Book ISBN : B999

Book not found.

Search Not Found

Enter your choice: 7

Enter Book Title : Java

Book not found.

Invalid Age

Enter your choice: 2

Enter Member ID : M003
Enter Name : Rahim
Enter Age : -5

Error: Age must be greater than 0.

Exit

==
LIBRARY MANAGEMENT SYSTEM
==

Enter your choice: 8

Thank you for using Library Management System.
Goodbye!

Validation Rules

- Book ISBN must be unique.
- Member ID must be unique.
- Age must be greater than 0.
- Book title cannot be empty.
- Author name cannot be empty.
- Prevent borrowing an already borrowed book.
- A member cannot borrow the same book twice.

Exception Handling

Handle the following errors using try-except:

- Invalid menu choice
- Invalid age input
- Empty input
- Duplicate ISBN

- Duplicate Member ID
- Unexpected runtime errors

The program should not crash because of invalid input.

GitHub Submission Rules

1. Create a public GitHub repository named:

python-library-management-system

2. Push the complete project to GitHub.

3. Submit your public github repository link

Submission: Submit the GitHub repository link before the deadline.